23Z711 - PROJECT WORK 1

UW-TIG: Uncertainty-Weighted Temporal Information Gain for Closed-Loop Defect Inspection

A Novel Active-Reinspection Planner and Its Simulation Testbed

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING
Branch : COMPUTER SCIENCE and ENGINEERING
of Anna University

NOVEMBER 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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ACKNOWLEDGEMENT

We would like to extend our sincere thanks to our Principal, Dr. K. Prakasan, Professor, for providing us with the opportunity to develop and complete our project in our field of interest.

We express our sincere thanks to our respectful Head of the Department, Dr. G. Sudha Sadasivam, Professor, for encouraging and allowing us to work on and present the project at our department for the partial fulfillment of the requirements leading to the award of the B.E. degree.

We take immense pleasure in conveying our thanks and a deep sense of gratitude to our faculty guide, Ms. L. Karthika, Department of Computer Science and Engineering, and our co-guide, Dr. N. Gopika Rani, for their relentless support, encouragement and suggestions, and for pushing us to test every claim against evidence rather than assumption throughout this project.

We also thank the authors of the open-source tools and public datasets on which this work stands: the PyBullet and gym-pybullet-drones simulators, the Ultralytics YOLO framework, and the MBDD2025 building-defect dataset. Finally, we thank our families and friends for their patience and support during the long experiment runs that this project required.

SYNOPSIS

Inspecting buildings, towers and pipelines for cracks, corrosion and leakage is repetitive work, and the autonomy that robots bring to it is usually limited to flying a fixed route or maximising how much surface is seen. Existing next-best-view planners decide where to look using geometry alone: they have no idea what a defect is, they forget everything between flights, and they never treat an unsure detection as a reason to look again. This project designs a planner that does.

Our contribution is **UW-TIG (Uncertainty-Weighted Temporal Information Gain)**. It keeps the cost-normalised information-gain formulation common to recent uncertainty-driven planners (Rückin et al., 2023; Moon et al., 2026) and adds a coverage guarantee, detection uncertainty from a trained YOLO detector, persistent cross-mission defect memory with growth tracking, a persistent-monitoring staleness term, and a two-step receding-horizon lookahead. Because a physical drone and a real defective structure are not needed to show or measure this, the second major contribution is a **reproducible simulation testbed**: an open-air utility yard in PyBullet textured with real photographs from the MBDD2025 dataset, a 54-viewpoint candidate graph, four defect scenarios, a fast kinematic mode for statistics and a PID-flight mode for demonstration, and an evaluation harness with separate development and test seeds.

On held-out test seeds the testbed reports UW-TIG at precision 0.949, recall 0.967 and F1 0.953, compared with five re-implemented published planners. It ties the strongest of them on detection quality (F1 0.950) while flying 39% less. The report also documents what did not work, what was wrong with our own measurements, and the limitations that remain.

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1. INTRODUCTION

Utility infrastructure – building facades, bridges, pipelines, transmission towers – degrades continuously. Cracks widen, corrosion spreads and leakage stains grow, and the only way to know is to look, repeatedly, over months and years. Unmanned aerial vehicles have made looking cheaper and safer, and deep-learning detectors can now find defects in the photographs they bring back. What has advanced much more slowly is the part in between: the decision of *where the vehicle should look next*.

In most deployed systems that decision is made before take-off. A route is planned, the drone flies it, and the images are analysed afterwards. Detection confidence never changes the route, a defect found last month is not remembered this month, and a blurry, low-confidence detection is either trusted or discarded rather than *checked*. Research on next-best-view (NBV) planning does close the loop between what has been seen and where to go, but it optimises coverage of geometry: the planner is rewarded for revealing unseen surface, whether or not that surface holds anything of interest.

This project asks a simple question: **what changes if the planner knows what a defect is, knows how sure the detector is, and remembers what it saw last time?** Answering it properly needs two things that are rarely built together – a planner with those three abilities, and an experimental environment in which the planner can be compared fairly against published alternatives, many times, with ground truth. This report is about both.

1.1. Objective

The objectives of the project are:

- To design a planner, UW-TIG, that selects inspection viewpoints using detector uncertainty, defect growth recorded in persistent memory, revisit staleness and a short lookahead, in addition to the geometric information gain used by the planners of recent work (Rückin et al., 2023).
- To build a simulation testbed in which this planner and several published planners can be run on identical scenes, detectors, seeds and scoring, and in which the defect appearance is real (photographs), not synthetic.
- To measure the planner honestly: choose every parameter on development seeds, report once on separate test seeds, include ablations of every added mechanism, and report failures alongside successes.

1.2. Overview

The system is a closed loop with five stages (Chapter 4). A **planner** chooses one of 54 candidate viewpoints. A **simulated camera** renders what the drone would see from there. A trained **YOLO detector** reports defects with a confidence and an uncertainty (the spread of its confidence over five photometrically perturbed copies of the frame). Each detection is **ray-cast** from the camera onto the known wall plane to obtain a 3-D position and a physical size. A **persistent memory** (PostgreSQL with pgvector, and Neo4j) matches the detection to a defect it has seen before, or registers a new one, and records how much the defect has grown. The updated uncertainty and growth feed back into the planner's next decision, and are re-loaded at the start of the next mission.

The simulation is as much a part of the contribution as the planner. It consists of an open-air utility yard with nine inspectable surfaces whose textures are cropped from real MBDD2025 photographs, a candidate-viewpoint graph with two standoff distances chosen from a detector diagnostic, four scenarios (static, uncertain, growing, multiple defects), and an evaluation protocol with a development/test split. Because the simulation is cheap, the planner comparison could be run over 5 test seeds × 4 scenarios × 12 planners × 3 missions – an experiment that would be impractical with a physical drone and that is, in any case, impossible to repeat identically in the field.

1.3. Scope

In scope: the UW-TIG planner and its ablations; re-implementations (at the level of the viewpoint-selection rule) of four published planners; the PyBullet testbed; a YOLO detector trained on a public dataset; the memory and geometry components that connect them; and a quantitative comparison with a statistical protocol.

Out of scope: hardware flight; obstacle avoidance and dynamics beyond a PID-controlled quadrotor model; photo-realistic rendering (Section 4.12 explains why this is a deliberate choice); and the authors' full original systems for the compared papers. These boundaries are repeated, with their consequences, in Chapters 6 and 7.

2. LITERATURE SURVEY

2.1. Literature Survey

2.1.1. Uncertainty-aware informative path planning (2023 – 2026)

Rückin et al. [1] plan UAV paths for active learning in semantic mapping, using a model's own epistemic uncertainty (MC-Dropout) to decide where to look, normalised by travel cost. The structure is the closest to UW-TIG's utility, but the uncertainty is spent to *improve the model* within one session, not to monitor a fixed defect set across sessions. Vashisth et al. [2] learn adaptive informative path planning with deep reinforcement learning on dynamic graphs; Moon et al. (IA-TIGRIS) [3] present a sampling-based planner that reuses earlier planning effort and adapts to updated belief maps, evaluated on multirotor and fixed-wing UAVs; Li et al. [4] select views under motion and measurement uncertainty for reconstructing moving objects. These works show the field moving toward learned and adaptive planners, but all of them reason about information gain over a generic belief, not about defects with an identity and a history.

2.1.2. Inspection-specific planning (2023 – 2025)

Dhami et al. introduced GATSBI [8, 9], an online algorithm that routes a UAV over target surfaces of a bridge with a generalised travelling-salesman formulation; the 2024 extension adds a defect detector. It is the closest published system to inspection-aware planning, but it plans within a single session and has neither a detection-uncertainty term nor memory across sessions. Pred-NBV and MAP-NBV [10] use a learned shape-completion model to guide views, an indirect form of uncertainty about geometry rather than about defects.

Petit and Lussier Desbiens [5] (MOAR) plan risk-aware, multi-objective paths for infrastructure inspection that adapt to weather and battery state; Svystun et al. [6] adapt trajectories to the pitch of wind-turbine blades using segmentation, reporting a 78% reduction in inspection time; Jin et al. [7] combine global coverage planning with local reactive re-planning for surface inspection in partially unknown indoor environments. None of these plans around detected defects, and none revisits a defect across missions. Two 2025 reviews [11, 12] of decision-making UAV path planning and of UAV infrastructure inspection both identify perception-driven flight planning, and fusion of detection with planning, as an open problem.

2.1.3. Classical building blocks

The recent planners above rest on older ideas, three of which UW-TIG also uses as ingredients. Isler et al. [23] formulate next-best-view selection as maximising an entropy-based information gain, normalised by travel cost; Bircher et al. [24] plan over a tree

of motions and execute only the first step (a receding horizon); Alamdari, Fata and Smith [25] minimise the maximum latency between observations for persistent monitoring. We re-implement these three selection rules as baselines so that the comparison spans both generations of planner, but they are not the reference point of this work: they have no notion of a defect, and the contribution is measured primarily against the recent, perception-aware planners of Sections 2.1.1 and 2.1.2.

2.1.4. Defect detection and datasets

Zha et al. [13] publish MBDD2025, 14,471 UAV photographs of building surfaces in six structure types and five defect classes (crack, leakage, abscission, corrosion, bulge), with bounding-box annotations and a benchmark of 35 detectors; it is the dataset used here. Recent detectors report high scores on narrower problems: Huang et al. [14] reach 0.913 precision, 0.766 recall and 0.864 mAP@50 for concrete cracks with an improved YOLOv11; Inam et al. [15] report 0.977 / 0.967 / 0.993 for bridge cracks (one class); Li, Shi and Sun [16] report 0.932 / 0.924 / 0.926 on a private 5,000-image utility-tunnel dataset. Shen et al. [17] propose TinyDef-DETR, a transformer detector for transmission-line defects, and Hebbache et al. [18] a saliency-guided YOLOX detector for bridge defects. These results are not directly comparable to a five-class public benchmark, a point Chapter 6 returns to.

2.1.5. Simulation platforms

AirSim [19] builds on Unreal Engine for high-fidelity rendering and physics; Flightmare [20] couples a Unity renderer with a separate fast physics engine; GRaD-Nav [21] represents the frontier with 3-D Gaussian Splatting. All three address the gap between rendered and real imagery when a model is *trained* on synthetic images. gym-pybullet-drones [22] provides a PyBullet quadrotor with PID control and an onboard camera, and is the physics layer used here.

Table 2.1 Summary of the reviewed planning literature

Paper, authors and year	Technique	Advantages	Limitations
Rückin et al., T-RO 2023 [1]	MC-Dropout uncertainty drives path	Real model uncertainty in planning	Single session; goal is retraining
Dhami et al. (GATSBI), 2023/24 [8,9]	GTSP routing + detector	Inspection-aware, cost-aware	Single session; no uncertainty/memory
Vashisth et al., 2024 [2]	Deep RL on dynamic graphs	Learned, adaptive	Needs training; no defect history
Petit & Lussier Desbiens, ICRA 2024 [5]	Risk-aware multi-objective planning	Weather/battery adaptive	No perception or defects
Svystun et al., 2024 [6]	Segmentation-guided trajectories	78% shorter inspection	One structure type; no revisits
Jin et al., 2025 [7]	Global coverage + local replanning	Handles unmapped obstacles	Coverage only
Moon et al., T-RO 2026 [3]	Incremental sampling-based IPP	Fast online replanning	Generic gain; no defects
Li et al., RSS 2026 [4]	NBV under motion uncertainty	Plans over uncertain futures	Reconstruction, not inspection
Isler / Bircher, ICRA 2016 [23,24]	Entropy-gain NBV; receding horizon	Classical building blocks (baselines)	Geometry only; no defects, no memory
Alamdari et al., IJRR 2014 [25]	Max-latency revisit scheduling	Guaranteed regular revisits	Time only; no perception

2.2. Inference

Three patterns emerge from the survey. First, **planning has become more capable but not more defect-aware**: 2023–2026 planners add learning, risk-awareness and incremental re-planning, yet their objective remains coverage or information gain over a generic belief. Second, **uncertainty-driven planning exists but is spent inside a single session** and aimed at improving a model, not at monitoring defects. Third, **persistence exists only as a revisit schedule**: Alamdari et al. know *when* to return, but nothing in the reviewed planners decides to return because of *what was seen*, such as a detection that was unsure or a defect that was growing.

On the detection side, the strongest published numbers come from single-class or private data, and the planning literature almost never evaluates against a real trained detector, relying instead on assumed ground-truth detections. A planner evaluated against a real detector inherits its failure modes – low recall at short range, false positives from neighbouring surfaces, poor calibration – and these turned out to matter greatly in this project (Chapters 5 and 6).

2.3. Research Gap

No reviewed work combines, in one planner that is evaluated against a real trained detector: (i) detection uncertainty as a reason to revisit; (ii) defect identity and growth remembered across missions; (iii) revisit staleness; and (iv) a cost-aware, short-horizon decision rule. Moreover, no reviewed work provides a controlled testbed in which such a planner can be compared with several published planners under identical conditions, with real defect imagery and a development/test split.

2.4. Research Objectives

- Introduce UW-TIG, a planner whose utility combines coverage information gain, detector uncertainty, defect growth, staleness and cost, selected with a two-step lookahead, preceded by an explicit coverage guarantee.
- Build a simulation testbed with real defect imagery, a fixed viewpoint graph, four scenarios and a leakage-free memory, so that every planner is measured on exactly the same inputs.
- Re-implement the selection rules of four published planners and compare them with UW-TIG and with random search.
- Ablate every added mechanism and report which ones measurably help, which do not, and why.
- Report limitations, including the calibration of the uncertainty signal and the failure of the growing-defect scenario.

3. SYSTEM REQUIREMENTS

3.1. Hardware Requirements

- **Development machine:** Windows 11 workstation, multi-core x86-64 processor (sweeps run several seeds in parallel).
- **Memory:** enough for several parallel seed processes, each holding a PyBullet client, a YOLO model and the detection memory.
- **Storage:** several GB for the dataset (MBDD2025, 14,471 images and labels), model weights, results and logs.
- **GPU (training only):** one NVIDIA L4 (24 GB), rented, used for the 100-epoch YOLOv8s training and for a later YOLO11m retrain. Simulation and evaluation run on CPU.

3.2. Software Requirements

Table 3.1 Software stack

Component	Tool / version	Purpose
Language	Python 3.12	All simulation, planning, perception and analysis code
Physics and rendering	PyBullet; gym-pybullet-drones	Quadrotor with PID control (DSLPIDControl), onboard camera, TinyRenderer
Detection	Ultralytics YOLOv8 / YOLO11; OpenCV; NumPy	Defect detector, test-time augmentation, image resampling
Persistent memory	PostgreSQL + pgvector; Neo4j	Detection log, appearance/spatial matching; defect identity graph
Deployment	Docker Compose	Runs both databases locally
Textures	Pillow	Wall, ground and sky texture generation and cropping
Analysis and reporting	matplotlib; ReportLab; pptxgenjs	Charts, this report, the review deck
Version control	Git / GitHub	Reproducibility of every reported number

4. SYSTEM DESIGN

4.1. Overall Architecture

Fig. 4.1 shows the closed loop. The planner, the simulated drone and camera, the detector, the geometry module and the persistent memory form a cycle; the feedback from memory to planner is what distinguishes this system from a pipeline that runs once. The same loop runs in two modes (Section 4.3): a fast kinematic mode used for the statistics and a PID-flight mode used for demonstration.

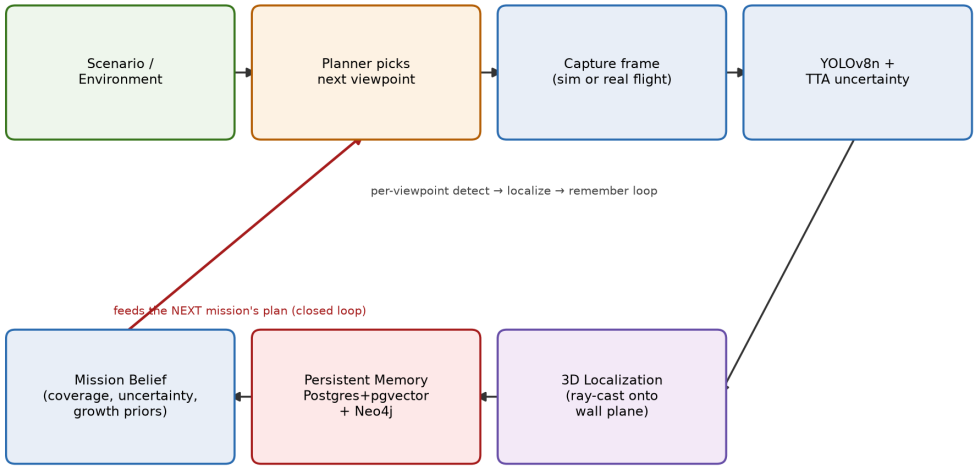


Fig. 4.1 System architecture: the five-stage closed loop

4.2. Simulation Environment

The environment is an **open-air utility yard** (Fig. 4.2). It replaced an earlier enclosed two-room house: in an enclosed building the drone never sees anything but walls, the sky is a white void, and the scene reads as a house tour rather than infrastructure. The yard has two freestanding buildings – Building A (four walls, 4 m × 4 m) and Building B (three walls, 2.6 m × 2.6 m) – about 11 m apart with open sky between them, and one inspection panel each on a decorative pipe rack and a lattice tower. In total there are **nine inspectable surfaces**. A sky backdrop with a skyline silhouette, a procedurally generated concrete ground texture and small props (crates, barrels) dress the scene.

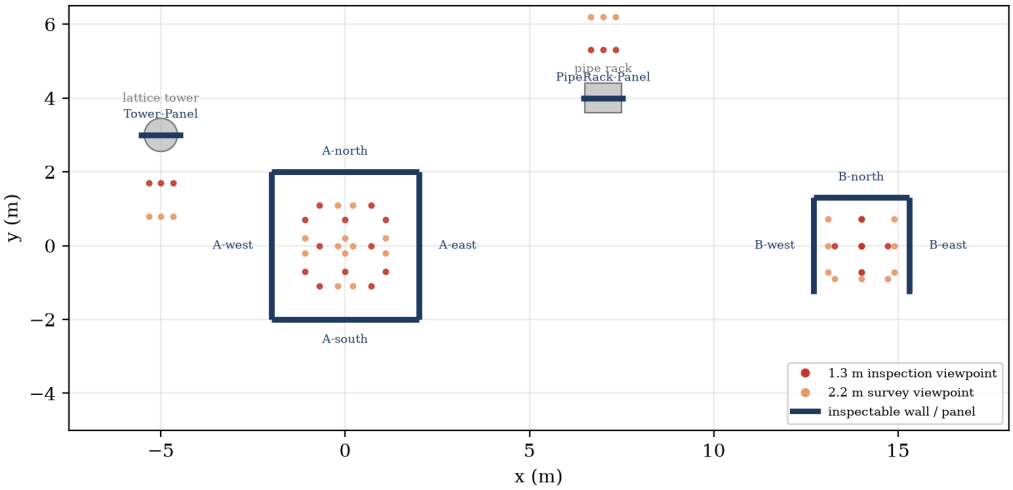


Fig. 4.2 Plan view of the yard: nine inspectable surfaces and the 54 candidate viewpoints

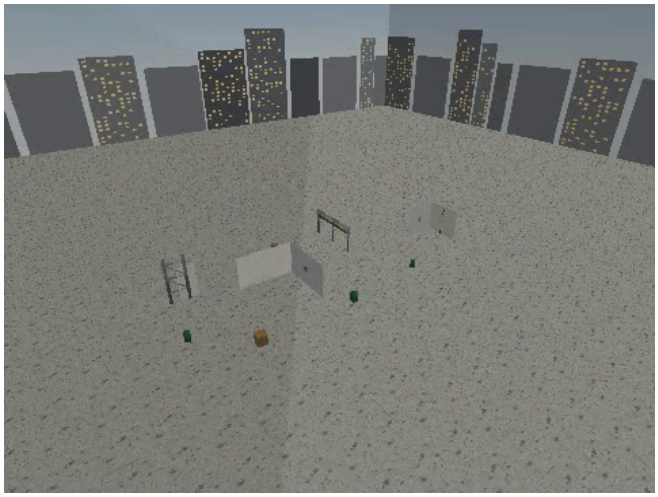


Fig. 4.3 Establishing shot of the simulated yard rendered with PyBullet

4.2.1. Walls as textured quads

Every inspectable surface is a flat quadrilateral mesh with a texture bound to it. A wall is defined by its centre, its yaw and its width; the plane it lies on, its outward normal and its in-plane axes follow from these (geometry.wall_plane). Using flat quads rather than boxes keeps the geometry of every wall analytically known, which is what makes ray-casting a detection back onto the wall exact (Section 4.7).

4.2.2. Real defect imagery as wall texture

A wall's defect is a real photograph from MBDD2025 applied as its texture. For a defect scenario, a photograph containing the wanted class is selected and a region is **cropped to the wall's true aspect ratio** (width divided by the 2.4 m wall height) at a long side of 1536 pixels. This crop was not in the first design: the early textures were square images stretched over walls up to 4 m wide, which distorted defects by roughly 2.2× and left them pixelated, so the detector was being tested on imagery it had never been trained for. Ground-truth defect centres are kept in a fixed 512-unit texture space so that the same physical spot is reproduced

from mission to mission and can be converted to a world position analytically (texture_uv_to_world).

Real photographs also contain defects of other classes: about 16% of MBDD2025 photos carry a second labelled defect. The scenario therefore records *which classes are present* in the cropped region, so that detecting a second real defect is not scored as a false positive (Section 4.10).

4.2.3. Camera and rendering

The onboard camera follows the gym-pybullet-drones convention: a 60° field of view rendered into a 320 × 240 frame with the projection matrix built for an aspect ratio of 1.0. The frame is therefore stretched horizontally by 4/3 relative to a square-pixel image. Because the detector was trained on undistorted photographs, every frame is resampled to square pixels before inference and the detected boxes are mapped back, so that localisation still uses the original geometry (Section 4.6).

Rendering uses PyBullet's TinyRenderer, a software rasteriser with shadow mapping. It is not a physically based renderer; the realism work in this project was limited to what changes the *detector's* input: consistent light direction and shadows, higher-resolution ground and sky textures, and a fix for a texture-tiling bug in box geometry (Section 5.2).

4.3. Two Simulation Modes

Detection and planning quality do not depend on rotor dynamics, so the statistical comparison does not need to step a physics engine for every flight. The testbed therefore has two modes that share the same scene, the same camera convention and the same planner code:

- **Kinematic mode** (experiments/kinematic_capture.py): the virtual camera is teleported to a candidate viewpoint and rendered. No PID stepping is performed. This mode runs the planner comparison.
- **PID-flight mode** (demo_uwtig_flight.py): a gym-pybullet-drones quadrotor flies between viewpoints under DSLPIDControl, with a smoothstep (ease-in, ease-out) reference to avoid velocity discontinuities, and a smoothed chase camera. This mode produces the demonstration videos.

The two modes use the same localisation mathematics, so what is verified in one carries to the other. The demonstration script accepts any planner by name, which allows the same scenario, seed and budget to be flown by different planners and the videos placed side by side – a way of *showing* the difference without any physical drone.

4.4. Candidate Viewpoint Graph

For each wall, the viewpoint generator creates stations at two **standoff distances**, 1.3 m and 2.2 m, and three lateral offsets (−0.55, 0 and +0.55 of the half-width), giving 54 viewpoints in all. Each viewpoint stores its pose and the **cells** of its wall that it can see. Surface coverage is discretised into cells of a fixed physical width of 1 m, so a 4 m wall has four cells and a 1.2 m panel one. This choice, following the fixed voxel *size* of classical volumetric planners [23], fixed a failure found early: with a fixed number of cells per wall, a narrow wall's single viewpoint saw all of its surface at once, earned a disproportionate information gain, and the planner froze on the small panels for an entire mission.

The 2.2 m standoff was added after a detection diagnostic on development seeds (Section 6.2). The cost of moving between viewpoints is the Euclidean distance plus a very small rotation term (1/3600 m per degree of yaw), whose only job is to break ties between viewpoints at exactly the same position, which occur in Building B because its 2.6 m width is exactly twice the 1.3 m standoff.

4.5. Scenarios

Table 4.1 The four defect scenarios

Scenario	Defects	Severity	Purpose
Static	A-west crack, A-east corrosion, B-north leakage (MBDD2025 photos)	1.0 (fixed)	Steady-state detection and coverage
Uncertain	A-south crack, B-east corrosion (MBDD2025 photos)	0.35 (faint)	Low-confidence, ambiguous defects
Multi-defect	Six walls, three classes (MBDD2025 photos)	0.9	Coverage breadth vs. reinspection depth
Growing	A-east corrosion, procedurally generated	0.4 to 1.3 over missions	Known-growth ground truth for the memory

Three scenarios use real photographs. The fourth, *growing*, needs a defect whose true size changes by a known amount between missions; no single-time-point photograph dataset can provide that, so it uses a procedural texture. This is the one scenario where the detector fails, for a reason documented in Section 6.7.

4.6. Perception

The detector is a YOLOv8s model at 640 px trained on MBDD2025 with a 70/20/10 split (Section 6.1). For each frame the system builds five inputs: the square-pixel frame and four photometric variants (brightness/contrast, additive noise, blur, and a blurred, re-brightened copy) that move no pixel geometry. The model runs on all five. For every detection in the original frame, the confidence of the best-overlapping same-class box in each variant (zero if there is none) is collected, and the **standard deviation** across the five is reported as the detection's **uncertainty**. This test-time-augmentation ensemble is a substitute for MC-Dropout; Section 6.6 and Chapter 7 discuss why a literal MC-Dropout was not

adopted.

4.7. 3-D Localisation

A detection's centre pixel is turned into a ray using the camera's view and projection matrices and intersected with the plane of the wall being inspected. A hit is accepted only if it lies **inside the wall's physical rectangle** (with a 5 cm margin). The rectangle test matters: a ray through a pixel that shows an adjacent wall, the ground or a prop still intersects the inspected wall's *infinite plane* somewhere, so without the test the defect was attributed to the wrong surface. On development seeds this accounted for 51–75% of all false positives, against one of 1,369 true positives. The box's two corners are localised the same way (without the rectangle test) to give a physical size in metres, used for growth.

4.8. Persistent Defect Memory

Memory has two stores. **PostgreSQL with pgvector** holds an append-only log of every detection and uses the stored appearance embedding of the cropped detection and its world position to match a new detection to an existing defect. **Neo4j** holds the defect graph: defect nodes linked to the missions that observed them and the walls they lie on. When a detection matches a known defect, its *growth* is the difference between its current and previous physical size; otherwise a new defect is registered. At the start of every mission after the first, the planner's belief is seeded with each active defect's last uncertainty and its growth. Memory keys are namespaced per scenario, planner and seed, so that no planner can see another's detections (Section 5.4).

4.9. The UW-TIG Planner

This section is the core of the contribution. The planner chooses the next viewpoint v from the 54 candidates. Its design has one front-end rule and one utility.

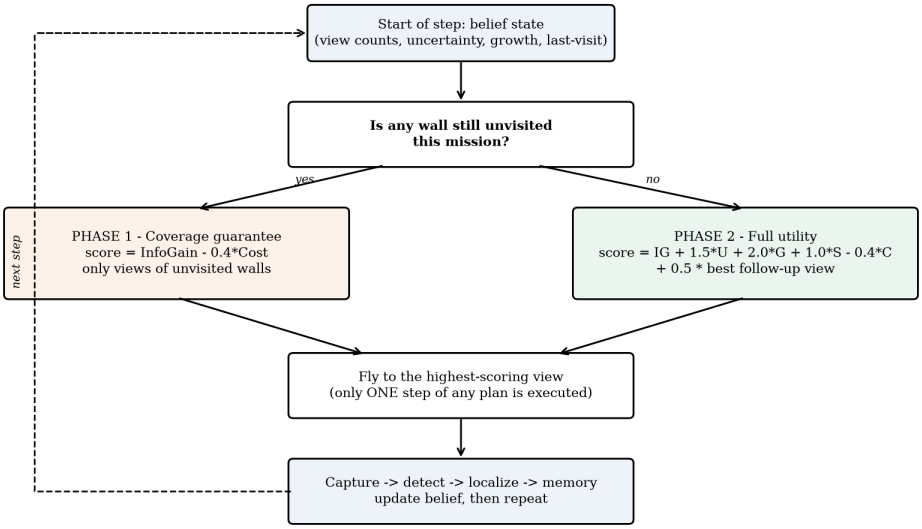


Fig. 4.4 UW-TIG decision flow for one step

4.9.1. The utility

$$\mathbf{U(v) = w_{ig}\cdot IG(v) + w_u\cdot Unc(v) + w_g\cdot Growth(v) + w_s\cdot Stale(v) - w_c\cdot Cost(v)}$$

with $w_{ig} = 1.0$, $w_u = 1.5$, $w_g = 2.0$, $w_s = 1.0$ and $w_c = 0.4$. The terms are as follows.

- **Information gain IG(v)** (the cost-normalised entropy term of [23], shared by the planners of Chapter 2). Each cell has a belief $p = 1 - 0.5\cdot e^{-0.6\cdot n}$ that it is already adequately observed after n views; its entropy is highest (1 bit) before any view and falls with each view. IG(v) is the sum of entropies of the cells the viewpoint can see.
- **Uncertainty Unc(v).** For each visible cell, the uncertainty last recorded there by the detector (Section 4.6), faded by $e^{-0.6\cdot n}$ so that re-viewing the same cell earns diminishing reward. This is what makes an unsure detection worth a second look.
- **Growth Growth(v).** The non-negative growth recorded in memory for the defects in view, with the same fading. This is the temporal term that lets a worsening defect attract the drone.
- **Staleness Stale(v).** $1 - e^{-0.15\cdot t}$ per cell, where t is the number of steps since the cell was last seen; it rises slowly, so a long-neglected area gradually becomes attractive (after the maximum-latency idea of Alamdari et al.).
- **Cost Cost(v).** Distance from the current viewpoint plus the small rotation term of Section 4.4.

4.9.2. The coverage guarantee

While any wall has not been visited *in this mission*, the planner restricts itself to the viewpoints of the unvisited walls and scores them with the plain information-gain rule alone, IG(v) − 0.4·Cost(v). Once every wall has been seen once, the full utility takes over. This was

added after a surprising observation: Isler-NBV and every early UW-TIG variant stopped at exactly 4 of 9 walls on every seed. Since a planner with *none* of the new terms hit the same ceiling, no re-weighting of the new terms could fix it. In the cost-normalised greedy rule, flying to the far building is never worth its cost against the information gain still available nearby. A weight is a fragile thing to hang a guarantee on; an explicit phase is not.

4.9.3. Two-step lookahead

In the full-utility phase, each candidate v_1 is scored as $U(v_1) + 0.5 \cdot \max_{v_2} U(v_2 \mid \text{after } v_1)$, where the second step uses the view counts that would result from visiting v_1 but the current uncertainty, growth and staleness (a future detection cannot be simulated). Only the first step is executed, after which the planner re-plans, in the manner of a receding-horizon planner [24].

4.9.4. Why the terms fade with repeated views

An early version let uncertainty and growth keep their full reward on every revisit. A real, persistent defect is re-detected on every look, so once found it earned its full reward again at no extra travel cost, and the lookahead exploited this so completely that flight distance collapsed to exactly 0.0 m for every UW-TIG variant: the planner camped on one viewpoint. Fading the terms with the view count, the same saturation used for coverage entropy, removed the exploit.

4.10. Reporting Rule and Metrics

A planner's output for a mission is the set of (wall, class) pairs it *reports*. A pair is reported once it has been seen in at least three separate looks (counted across missions, since memory carries the count) or once at confidence ≥ 0.45; the rule is identical for every planner and was chosen on development seeds by planner-averaged F1. Precision, recall and F1 are computed over these reports against the scenario's ground truth, with classes that are really present in the photograph but are not the scenario's primary class *ignored* (as in COCO-style evaluation). Other metrics are mean localisation error (metres), total flight distance, wall coverage, reinspection rate, expected calibration error (ECE) of the detector's confidence, and an uncertainty gap (mean uncertainty on false positives minus on true positives). A "legacy" precision that divides unique true positives by every raw false-positive box is also kept, to show how much of the improvement is a change in measurement (Section 6.2).

4.11. Re-implemented Baselines

Four published planners were added to the registry, each adapting the paper's viewpoint-selection rule to the same viewpoint graph and belief. They are faithful to the *rule*, not to the authors' full systems. Their parameters were set once from the papers' formulations and not tuned on any result.

Table 4.2 Re-implemented selection rules

Registry name	Paper	Selection rule
bircher_rhnbv	Bircher et al. 2016	Best depth-2 branch of gain × $e^{-0.5 \cdot cost}$; execute the first step only
gatsbi_gtsp	Dhami et al. (GATSBI)	Nearest-neighbour + 2-opt tour over the least-inspected walls, followed in order
ruckin_ipp	Rückin et al. 2022–23	Acquisition Σ(uncertainty, 0.5 if unobserved) / (1 + visit count), single-session belief
alamdari_latency	Alamdari et al. 2014	Greedy on maximum time since last visit
isler_nbv / random	Isler et al. 2016 / none	Cost-normalised entropy gain over unvisited views / uniform random

4.12. Why PyBullet and Not Unity or Unreal

A natural question is whether a game-engine or Gaussian-splatting renderer would make the simulation more convincing. A literature check suggests the answer is no *for this project*. AirSim [19], Flightmare [20] and GRaD-Nav [21] exist to close the gap between rendered and real imagery when a model is trained on synthetic images. Here the detector is trained and tested only on real photographs, applied as textures; the camera stays tightly framed on the wall under inspection and never sees the yard dressing. A better renderer would make the simulator prettier to watch but would not change one pixel that the detector learns from. PyBullet was therefore kept: it is scriptable from a terminal, runs headless in parallel, and gym-pybullet-drones already supplies PID flight. The honest limits – a software rasteriser, no sensor noise model, no wind – are listed in Section 6.8.

5. SYSTEM IMPLEMENTATION

5.1. Code Organisation

Table 5.1 Repository layout

Module	Responsibility
sim_house.py; scene/environment.py	Wall layout, textured quads, ground, sky, pipe rack, tower, props
experiments/kinematic_capture.py	Teleport-and-render camera for the statistical sweeps
demo_uwtig_flight.py; make_comparison_video.py	PID-flight demo; side-by-side planner videos
datagen/mbdd_textures.py; prepare_mbdd.py; oversample_bulge.py	MBDD2025 preparation, aspect-correct wall textures, class oversampling
perception/ml_detector.py; train_yolo.py; train_yolo_gpu.py	Detector with TTA uncertainty; CPU and GPU training
geometry.py	Camera matrices, pixel rays, wall planes, on-wall localisation
memory/store.py	PostgreSQL/pgvector log, Neo4j graph, identity matching and growth
planning/viewpoints.py; planning/planners.py	Viewpoint graph and cells; belief, UW-TIG, ablations and baselines
experiments/scenarios.py; mission.py; evaluate.py	Scenarios; the mission loop; scoring and the sweep driver
experiments/diagnose_detection.py; tune_report_rule.py	Planner-independent detector diagnostic; report-rule tuning on dev seeds

5.2. Implementing the Simulation

Walls. A single quad mesh (scene/wall_quad.obj) is scaled and rotated per wall, with its UV mapping chosen so that a texture pixel maps to a known world point; this analytic inverse was validated against the ray-plane path to under 2 cm for a known marker, and is used to score localisation error. **Lighting.** The capture call uses a fixed light direction with shadows enabled so that shading is consistent from view to view. **Textures.** Ground (concrete) and sky (gradient with a skyline silhouette) are generated procedurally at higher resolution than the first version; the sky is a backdrop around the yard so that the horizon is no longer a blank white void.

A tiling bug. A texture-tiling bug in PyBullet's box geometry (GEOM_BOX) was found during the realism pass and fixed. **A resolution hypothesis that was wrong.** Because early recall was low, the first suspicion was image resolution. A higher camera and inference resolution was tried and *reduced* recall (0.42 to 0.22). The causes turned out to be texture distortion and viewing distance (Section 6.2), which would not have been found if the hypothesis had been accepted without measurement.

5.3. The Mission Loop

Algorithm 1 summarises one mission (experiments/mission.py). The scenario sets the wall textures for this mission index and supplies ground truth; the belief is seeded from memory if the mission index exceeds one; then, for each of the 16 steps, the planner chooses a viewpoint, the camera captures it, the detector runs, and each detection is localised, matched in memory, and written back into the belief.

```
Algorithm 1: one mission
  belief <- MissionBelief(walls)
  if mission > 1: belief.seed_from_memory(run_key)
  scenario.setup_mission(mission)          # textures + ground truth
  for step in 1..16:
      v   <- planner.select_next(belief, current)
      img <- camera.capture(v)
      for d in detector(img):              # class, confidence, uncertainty
          xyz  <- localize_on_wall(d.center, pose, wall)   # reject if off-wall
          size <- distance(localize(d.corner1), localize(d.corner2))
          id, growth <- memory.match_or_create(run_key, wall, d, xyz, size)
          belief.record_detection(wall, xyz, d.uncertainty, growth)
      belief.record_visit(v);  current <- v
```

5.4. Experiment Harness and Protocol

- **Development / test split.** Every design choice (viewpoint standoffs, reporting rule, fixes) was made on development seeds 100–102. The reported numbers come from one run on test seeds 0–4 that was not tuned on.
- **Sweep size.** 4 scenarios × 12 planners × 5 seeds × 3 missions, 16 viewpoints per mission; results are averaged over static, uncertain and multi-defect (growing is reported separately).
- **Isolation.** Memory keys are namespaced as *scenario*|*planner*|*seed*. An early sweep let planners' detection histories leak into each other because identity matching was not scoped; this was caught by a targeted test and fixed before any reported number was produced.
- **Process isolation.** A single long process running about 140 mission-runs crashed three times with three different symptoms (allocation failure, silent exit, segmentation fault), consistent with PyBullet's software renderer accumulating resources over many render sessions. Each seed now runs in its own subprocess and the per-seed CSVs are merged (run_full_sweep.sh).

5.5. Debugging Episodes

Four defects in the work itself shaped the final design, and are recorded here because each changed a conclusion.

- **Camping.** The lookahead exploited non-decaying uncertainty/growth rewards (flight distance 0.0 m). Fixed by fading the terms with view count (Section 4.9.4).

- **Free spinning.** Building B's coincident viewpoints made turning in place free. An over-large rotation cost fixed it but collapsed Isler-NBV's coverage from 0.875 to 0.333; the final constant (1/3600 m per degree) breaks only exact ties.
- **Cells.** A fixed cell count per wall froze the planner on narrow panels; fixed physical cell width fixed it (Section 4.4).
- **Wrong-surface hits.** Ray hits outside the wall's rectangle were blamed on the inspected wall (Section 4.7).

5.6. Demonstration and Comparison Videos

demo_uwtig_flight.py flies a chosen planner under real PID control through three missions of the multi-defect scenario, writing a third-person flythrough and an onboard-camera video with detections, confidences and uncertainties overlaid. make_comparison_video.py splices the flythrough or inspection videos of different planners frame-by-frame, because the step count depends on the budget and mission count, not on the planner. Fig. 5.1 is one such frame.



Fig. 5.1 Same scenario, seed and timestep, three planners (left to right: UW-TIG, Isler-NBV, Random)

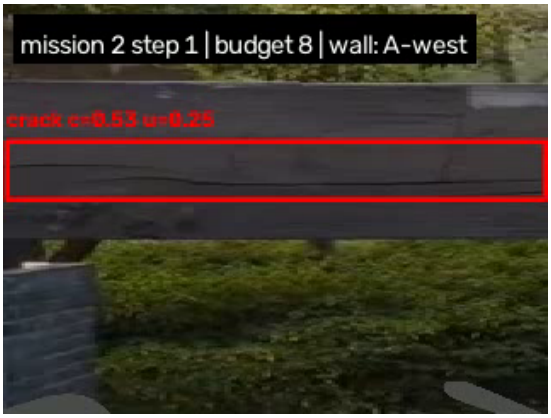


Fig. 5.2 Onboard camera frame from the demonstration: real corrosion detections with confidence and uncertainty

6. EXPERIMENTAL RESULTS

6.1. Detector Benchmark

The detector is evaluated on the held-out MBDD2025 test split of 1,448 images. Two checkpoints were trained: a CPU baseline (YOLOv8n, 320 px, 30 epochs) and the default GPU model (YOLOv8s, 640 px, 100 epochs, with the rarest class, bulge, oversampled about four times). A later YOLO11m retrain on the same GPU was evaluated but not adopted.

Table 6.1 Detector results on the held-out test set (1,448 images)

Model	mAP50	mAP50-95	Precision	Recall
YOLOv8n, 320 px, 30 epochs (CPU)	0.685	0.347	0.804	0.612
YOLOv8s, 640 px, 100 epochs (default)	0.884	0.506	0.874	0.842
YOLO11m, 640 px (retrain; not adopted)	0.890	0.510	0.869	0.863

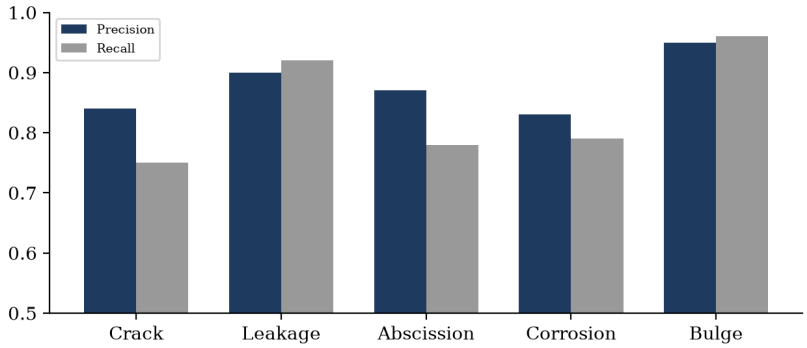


Fig. 6.1 Per-class precision and recall of the default detector

Crack is the hardest class (recall 0.75): cracks are thin, low-contrast features that lose most of their signal at low resolution, which is why moving from 320 px to 640 px nearly doubled crack recall. Bulge, the best class (0.95 / 0.96), was the rarest in the data and was oversampled. The default detector does *not* reach 0.90 on mean precision or recall; all the above-0.90 numbers in this report are closed-loop report-level metrics (Section 6.3), not detector metrics. Compared with the literature, this detector is below the single-class and private-data scores of Section 2.1.4 and above the recall of Huang et al.'s crack detector (0.766) on a harder five-class public task.

6.2. How the Closed-Loop Numbers Reached 0.90

The first complete sweep gave UW-TIG 0.76 precision, 0.63 recall and 0.65 F1. Rather than tune the planner, we asked whether the *measurement* was right. Five problems were diagnosed on development seeds 100–102, each affecting every planner equally:

Table 6.2 The five fixes

#	Problem	Fix
1	Precision divided unique true positives by every raw false-positive box, penalising reinspection itself	Score per inspection report (3 looks across missions, or one at confidence ≥ 0.45)
2	16% of photos hold a second real defect; detecting it counted as a false positive	Ignore classes present in the photo, COCO-style
3	Textures stretched ~2.2× and pixelated	Aspect-correct crop at 1536 px; square-pixel frames
4	Single-look recall 0.47 at 0.8 m, 0.85 at 1.3 m, 0.97 at 1.8 m, 0.99 at 2.2 m	Replace the 0.8/1.3 m tiers with 1.3 m and 2.2 m
5	Hits outside the wall rectangle blamed on the inspected wall (51–75% of false positives)	Reject hits off the wall (+5 cm)

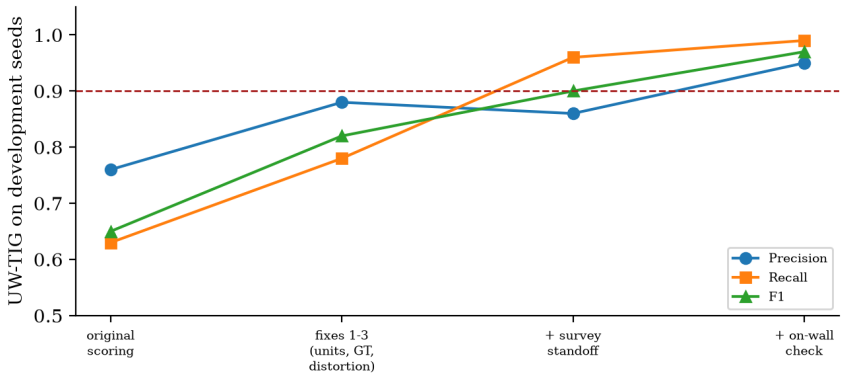


Fig. 6.2 UW-TIG on development seeds as the fixes were applied

Development progression (precision / recall / F1): original 0.76 / 0.63 / 0.65; fixes 1–3 0.88 / 0.78 / 0.82; plus the survey standoff 0.86 / 0.96 / 0.90; plus the on-wall check 0.95 / 0.99 / 0.97. The honest reading is that **a large part of the gain is measurement, not planning**: under the old scoring formula the final UW-TIG precision on test seeds would be 0.668.

6.3. Closed-Loop Comparison and Ablations

Table 6.3 UW-TIG and its ablations on test seeds 0–4 (mean over static, uncertain, multi-defect; confirmed reports)

Planner	Precision	Recall	F1	Loc. err (m)	Flight (m)	Coverage
UW-TIG (full)	0.949	0.967	0.953	0.70	125.1	1.00
no uncertainty term	0.949	0.967	0.953	0.71	124.6	1.00
no temporal (growth) term	0.949	0.967	0.953	0.70	125.1	1.00
no staleness term	0.967	0.956	0.958	0.57	80.6	1.00
no lookahead	0.967	0.956	0.958	0.59	85.2	1.00
no coverage phase	0.911	0.556	0.680	0.81	13.4	0.44

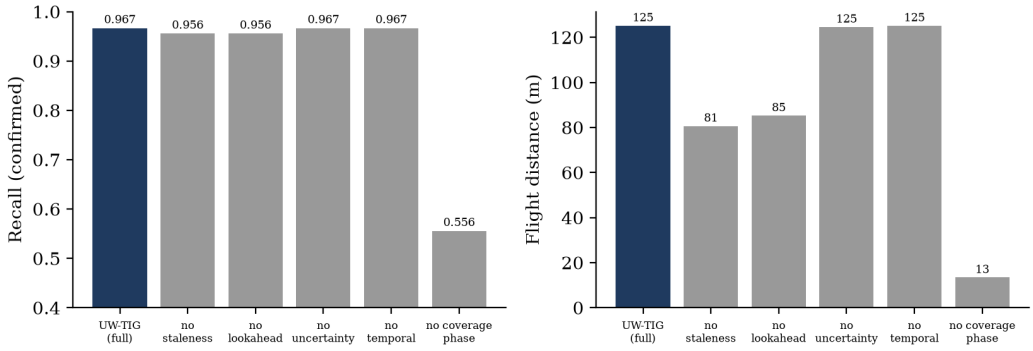


Fig. 6.3 Recall and flight distance of the ablations

UW-TIG reaches precision 0.949, recall 0.967 and F1 0.953. By scenario (precision / recall / F1) it scores static 0.964 / 0.933 / 0.940, uncertain 0.893 / 0.967 / 0.925 and multi-defect 0.989 / 1.000 / 0.995; the per-seed F1 values are 0.943, 0.890, 1.000, 1.000 and 0.933. **The averages clear 0.90, but not every slice does**: the deliberately faint uncertain scenario is at 0.893 precision and one test seed is at F1 0.890.

The **coverage guarantee is the decisive component**: without it recall falls from 0.967 to 0.556 and F1 from 0.953 to 0.680, with coverage back at 0.44 (4 of 9 walls). Two ablations are slightly *better* than the full planner: without staleness or without the lookahead, precision is 0.967, localisation error 0.57–0.59 m and flight 32–36% shorter, because once a single survey look at 2.2 m is about 99% reliable, revisiting for staleness or planning ahead adds flying but not accuracy.

6.4. Why Removing Uncertainty or Growth Changes Nothing

Table 6.3 shows that the no-uncertainty and no-growth ablations are identical, or nearly so, to the full planner. We checked the code rather than assume, and found two reasons.

- **The coverage phase never evaluates these terms.** While any wall is unvisited, the planner scores only IG − 0.4·Cost. With nine walls and a 16-viewpoint budget, roughly half of every mission is spent there, and the three missions of a seed each re-run this phase because it is defined per mission.
- **Growth is near zero by construction in these scenarios.** Growth is the change in a defect's measured size, clipped to be non-negative; static, uncertain and multi-defect defects never grow, so growth is only detector noise around a constant. Removing the term leaves flight distance bit-identical (125.1 m).
- **Uncertainty is real but small.** The TTA spread is typically 0.1–0.3, so 1.5× that is small beside the other terms. Removing it shifts the path slightly (124.6 m against 125.1 m) but not how many looks each wall gets, which is all that confirmed precision and recall depend on.

This is an open finding, not a vindication: it means that in the current configuration two of the four novel mechanisms do not yet influence the outcome, and that re-weighting them, or testing on scenarios with real growth, is the next step (Chapter 7).

6.5. Head-to-Head with Published Planners

Table 6.4 Planner comparison on test seeds 0–4 (confirmed reports)

Planner	Prec.	Recall	F1	F1 1-look	Loc. err (m)	Flight (m)	Cov.
Random baseline	0.921	0.837	0.873	0.800	0.85	394.9	1.00
Isler-NBV (classical, 2016)	0.890	0.556	0.671	0.649	0.94	30.7	0.44
Bircher et al. 2016	0.917	0.756	0.814	0.814	0.80	47.9	0.67
Dhami et al. (GATSBI)	0.887	0.922	0.900	0.846	0.63	125.4	1.00
Alamdari et al. 2014	0.924	0.967	0.941	0.909	0.73	167.7	1.00
Rückin et al. 2023	0.959	0.944	0.950	0.931	0.86	204.3	1.00
UW-TIG (ours)	0.949	0.967	0.953	0.939	0.70	125.1	1.00

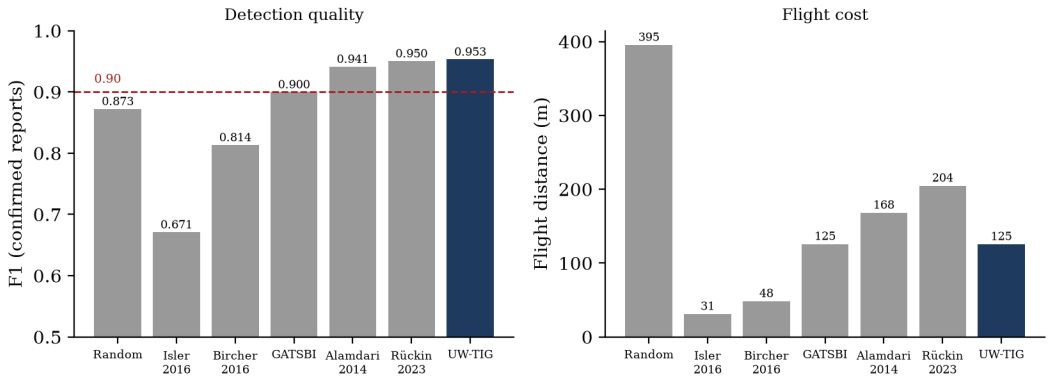


Fig. 6.4 F1 and flight distance of all compared planners

UW-TIG ties the strongest planner, Rückin's IPP, on detection quality (F1 0.953 against 0.950, well inside the seed-to-seed spread, UW-TIG's own per-seed F1 spanning 0.890 to 1.000). **Where it differs is cost**: 125.1 m of flight against 204.3 m, 39% less, with the lowest localisation error of the seven planners. Rückin's rule has slightly higher precision (0.959 against 0.949), which supports uncertainty-driven acquisition as a strong rule. The published rules that fail (Isler, Bircher) fail on *coverage*, 0.44 and 0.67 of the walls, the same cost-scale ceiling the coverage phase removes. These are selection-rule re-implementations, not the authors' systems, and no number from the original papers is reproduced or claimed.

6.6. Calibration of the Uncertainty Signal

The study of Rückin et al.'s evaluation prompted a check we had not run on ourselves: is the uncertainty UW-TIG weights informative, and is the detector's confidence calibrated? Across all twelve planners the expected calibration error is 0.29–0.33, far above a calibrated

model, and the uncertainty gap is *negative* for every planner (−0.12 to −0.08): uncertainty is on average slightly *lower* on false positives than on true positives. A plausible, unconfirmed explanation is that real, hard-to-see defects sit near the decision boundary and vary more under photometric perturbation than a consistently mis-recognised texture. The signal is used only to rank views within a mission, never as a probability, so the measured precision and recall stand; but the claim that high uncertainty means "likely wrong" is not supported.

We also checked whether switching to MC-Dropout is a simple fix. In the installed Ultralytics, stock YOLOv8/v11 detection models contain no dropout layer anywhere (it exists only in the classification head). A true MC-Dropout would need dropout spliced into the neck or head through a custom model definition, a full retrain, and has no guarantee of better calibration, since dropout variance is known to track correctness poorly on confident, well-converged detectors. It is therefore listed as future work.

6.7. The Growing Scenario

In the growing scenario the detector recalls essentially nothing. Its defect is a procedural corrosion texture, and a detector trained only on real photographs does not recognise it (zero detections above confidence 0.1 on the rendered frame, checked directly). The memory mechanism itself was verified in isolation – a defect re-detected at a shifted position and larger size was matched and its growth computed, while a distant detection created a new identity – so the failure is the detector's domain generalisation, not the memory. For this reason the growing scenario is excluded from the comparison tables and the growth term has not yet been shown to help.

6.8. Threats to Validity

- **Selection-rule baselines.** The compared planners are re-implemented rules, not full systems.
- **Five seeds.** Enough to see the coverage effect, not enough to separate the staleness, lookahead, uncertainty and growth ablations.
- **Measurement versus planning.** Five fixes in measurement raised every planner; the improvement over the first sweep should not be read as a planning gain.
- **Simulation realism.** A software rasteriser, flat-quad walls, no wind or sensor noise, a kinematic mode without dynamics; absolute detector scores would differ outdoors.
- **Single-time-point data.** MBDD2025 has no growth sequences, so temporal claims rest on one procedural scenario the detector cannot see.

7. FUTURE SCOPE AND ENHANCEMENTS

- **Make the novel terms matter.** Re-weight or normalise the uncertainty and growth terms, and evaluate on scenarios with real growth, so that the ablations in Section 6.4 separate.
- **Better uncertainty.** Try post-hoc temperature scaling first; then prototype MC-Dropout by splicing dropout into the detection head and retraining.
- **Close the growing-defect gap.** Mix synthetic defect imagery into training, or obtain a longitudinal real dataset (the SDNET2018 crack augmentation was prepared but its download is bot-protected).
- **Lift the weak slices.** Raise the uncertain scenario's precision above 0.90 through scenario-specific confirmation-rule tuning, and narrow Rückin's precision lead by re-weighting utility terms.
- **Fuller baselines.** Re-implement the compared papers, and IA-TIGRIS (2026), as full systems with sampling-based planning and online retraining.
- **More seeds** for the ablations, and a decision whether staleness and lookahead stay on by default.
- **Richer simulation.** Pipe segments and truss rigs as first-class inspectable geometry; sensor noise and wind; a Gaussian-splatting or game-engine renderer if the work moves to training on synthetic imagery.
- **A stronger detector.** The YOLO11m retrain (mAP50 0.890, precision 0.869, recall 0.863) is a marginal, mixed gain; adopting it requires re-running the sweep. YOLOv8m/l and 1280 px were not tried.

8. CONCLUSION

This project built and measured a planner that knows what a defect is. UW-TIG combines an information-gain term with a coverage guarantee, detector uncertainty, cross-mission defect memory with growth, revisit staleness and a two-step lookahead, and was evaluated on a simulation testbed built for the purpose. On held-out test seeds it reaches precision 0.949, recall 0.967 and F1 0.953, above 0.90 on all three, which neither the classical information-gain baseline nor random search reaches, and it ties the strongest of five re-implemented published planners while flying 39% less.

Equally important is what the measurements say about the work. The coverage guarantee, not the more elaborate terms, accounts for the large gain over the plain information-gain baseline; the uncertainty and growth terms do not yet change any reported number; much of the improvement over the first sweep came from correcting five flaws in how the system was *measured*; the uncertainty signal is poorly calibrated; and the growing-defect scenario fails because of the detector, not the memory. We report these because a planner that is only shown where it wins cannot be trusted where it matters.

The simulation testbed is a result in its own right. It gives a controlled, repeatable setting – real defect imagery, a fixed viewpoint graph, leakage-free memory, a development/test split and a one-command sweep – in which any planner can be added to the registry and compared at the cost of a few CPU-hours, without a drone. We expect it to be the more reusable part of this work.

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APPENDICES

Appendix A - UW-TIG planner (planning/planners.py)

```
class UWTIGPlanner(BasePlanner):
    """Novel: Uncertainty-Weighted Temporal Information Gain. Strictly
    generalizes IslerNBVPlanner by adding defect-uncertainty, temporal-
    growth, and persistent-monitoring staleness terms sourced from
    persistent memory, and -- unlike the coverage-only baselines -- is
    allowed to revisit an already-inspected viewpoint when that utility
    gain is high (active reinspection). Selects via a 2-step
    receding-horizon lookahead rather than pure 1-step greedy (see module
    docstring for the literature this is adapted from and its caveats).

    Also runs a **coverage-guarantee phase**: while any wall has zero visits
    this mission, `select_next` restricts itself to Isler-NBV's own
    formulation (ig - cost, no uncertainty/temporal/staleness, no lookahead)
    over ONLY the still-unvisited walls' viewpoints. This was added after
    finding that `coverage_frac` was stuck at exactly 0.333 (3 of 9 walls)
    for EVERY UW-TIG ablation *and* for `isler_nbv` itself, identically,
    across every seed -- i.e. this is not a side effect of the
    uncertainty/temporal/staleness weights (a weight retune can't reliably
    fix it), it's the cost-normalized-greedy formulation itself finding
    that, on this viewpoint graph, traveling to the far building is never
    worth its cost relative to the coverage-entropy reward available,
    regardless of weighting. A weight scale is fragile to depend on for a
    coverage guarantee; an explicit phase is not. Once every wall has been
    visited at least once this mission, control passes to the full
    multi-term utility below for the remaining budget -- so this only
    changes *when* full coverage happens, not whether UW-TIG's reinspection
    behavior still runs (as long as budget > wall count, which it is
    throughout this project's evaluation). Ablation: `uwtig_no_coverage_first`
    reproduces the original (recall-limited) behavior for comparison."""
    name = "uwtig"
    w_ig = 1.0
    w_cost = 0.4
    w_uncertainty = 1.5
    w_temporal = 2.0
    w_staleness = 1.0
    lookahead_discount = 0.5  # weight on the best-achievable second step
    coverage_first = True

    def _utility(self, belief, from_id, v, view_count=None):
        vc = belief.view_count if view_count is None else view_count
        ig = self.w_ig * sum(
            _entropy(1 - 0.5 * math.exp(-VIEW_SATURATION_K * vc[(v.wall, c)]))
            for c in v.visible_cells)
        u = self.w_uncertainty * belief.uncertainty_score(v)
        t = self.w_temporal * belief.temporal_score(v)
        s = self.w_staleness * belief.staleness_score(v)
        cost = self.w_cost * self._cost(from_id, v)
        return ig + u + t + s - cost

    def _view_count_after(self, belief, v):
        """What-if view_count if `v` were visited next -- used only to give
        the lookahead's second step a coverage-aware (not stale) picture of
        information gain. Cannot simulate a future detector output, so
        uncertainty/temporal/staleness are read from the current belief for
        the second step (see module docstring caveat)."""
        vc = dict(belief.view_count)
        for c in v.visible_cells:
            vc[(v.wall, c)] = vc.get((v.wall, c), 0) + 1
        return vc

    def _unvisited_walls(self, belief):
        visited_walls = {self.by_id[vid].wall for vid in belief.visited_ids}
        return [w for w in self.all_walls if w not in visited_walls]

    def select_next(self, belief, current_id):
        if self.coverage_first:
            unvisited_walls = self._unvisited_walls(belief)
            if unvisited_walls:
                pool = [v for v in self.viewpoints if v.wall in unvisited_walls]
                scored = [(self.w_ig * belief.information_gain(v) - self.w_cost * self._cost(current_id, v), v)
                          for v in pool]
                return max(scored, key=lambda t: t[0])[1]
        best_v, best_val = None, -math.inf
```

```
        for v1 in self.viewpoints:
            val1 = self._utility(belief, current_id, v1)
            if self.lookahead_discount:
                vc_after = self._view_count_after(belief, v1)
                best_second = max(self._utility(belief, v1.id, v2, view_count=vc_after)
                                  for v2 in self.viewpoints)
                val1 += self.lookahead_discount * best_second
            if val1 > best_val:
                best_val, best_v = val1, v1
        return best_v
```

Appendix B - Mission belief: information gain, uncertainty, growth, staleness (planning/planners.py)

```
class MissionBelief:
    """Per-mission state: coverage (geometry-only) + defect uncertainty/growth
    (defect-aware), both keyed by (wall_name, cell_index). Takes the actual
    wall dicts (not just names) since cell count is per-wall, proportional
    to physical width -- see planning.viewpoints.n_cells_for_wall."""

    def __init__(self, wall_segments):
        cells = {w["name"]: n_cells_for_wall(w) for w in wall_segments}
        self.view_count = {(w, c): 0 for w, n in cells.items() for c in range(n)}
        self.uncertainty = {(w, c): 0.0 for w, n in cells.items() for c in range(n)}
        self.growth = {(w, c): 0.0 for w, n in cells.items() for c in range(n)}
        # In-mission revisit latency (steps since last view), not a
        # cross-mission real-time latency -- see UWTIGPlanner's docstring.
        self.last_visit_step = {(w, c): -1 for w, n in cells.items() for c in range(n)}
        self.step = 0
        self.visited_ids = []

    def seed_temporal_priors(self, priors):
        """priors: dict[(wall, cell)] -> (uncertainty, growth), typically
        built by experiments/mission.py from memory.store history at the
        start of a mission -- this is the "prior inspections inform this
        mission's plan" temporal-intelligence link."""
        for key, (u, g) in priors.items():
            if key in self.uncertainty:
                self.uncertainty[key] = u
                self.growth[key] = g

    def coverage_entropy(self, wall, cell):
        # p = belief the cell is ALREADY adequately observed. Starts at 0.5
        # (maximally uncertain -> max entropy -> max info gain from a first
        # look) and climbs toward 1 (certain, resolved) with more views.
        p = 1 - 0.5 * math.exp(-VIEW_SATURATION_K * self.view_count[(wall, cell)])
        return _entropy(p)

    def record_visit(self, viewpoint):
        self.visited_ids.append(viewpoint.id)
        for c in viewpoint.visible_cells:
            key = (viewpoint.wall, c)
            self.view_count[key] += 1
            self.last_visit_step[key] = self.step
        self.step += 1

    def record_detection(self, wall, world_xyz, uncertainty, growth, wall_dict):
        cell = cell_index_for_world_point(wall_dict, world_xyz)
        key = (wall_dict["name"], cell)
        self.uncertainty[key] = uncertainty
        self.growth[key] = max(0.0, growth)

    def information_gain(self, viewpoint):
        return sum(self.coverage_entropy(viewpoint.wall, c) for c in viewpoint.visible_cells)

    def _repeat_decay(self, wall, cell):
        # Diminishing marginal value of acting on the SAME reading again --
        # keyed by view_count (times this cell has actually been looked at),
        # not by "since the detector last reported something," because a
        # real, persistent defect gets re-detected on every single look: a
        # since-last-update decay never fires for it and the planner camps
        # on that one viewpoint forever once it finds anything (caught via
        # the lookahead below making that camping total, flight_dist ->
        # exactly 0.0, in eval). Reusing coverage_entropy's saturation here
        # extends the same "repeated identical observation earns
        # diminishing reward" logic from coverage to uncertainty/growth.
        return math.exp(-VIEW_SATURATION_K * self.view_count[(wall, cell)])

    def uncertainty_score(self, viewpoint):
```

```
        return sum(self.uncertainty[(viewpoint.wall, c)] * self._repeat_decay(viewpoint.wall, c)
                   for c in viewpoint.visible_cells)

    def temporal_score(self, viewpoint):
        return sum(self.growth[(viewpoint.wall, c)] * self._repeat_decay(viewpoint.wall, c)
                   for c in viewpoint.visible_cells)

    def staleness(self, wall, cell):
        last = self.last_visit_step[(wall, cell)]
        elapsed = self.step if last < 0 else self.step - last
        return 1 - math.exp(-STALENESS_SATURATION_K * elapsed)

    def staleness_score(self, viewpoint):
        return sum(self.staleness(viewpoint.wall, c) for c in viewpoint.visible_cells)
```

Appendix C - Ablation planners (planning/planners.py)

```
class UWTIGNoUncertaintyPlanner(UWTIGPlanner):
    """Ablation: temporal + staleness + coverage, no detection-uncertainty term."""
    name = "uwtig_no_uncertainty"
    w_uncertainty = 0.0

class UWTIGNoTemporalPlanner(UWTIGPlanner):
    """Ablation: uncertainty + staleness + coverage, no temporal-growth term."""
    name = "uwtig_no_temporal"
    w_temporal = 0.0

class UWTIGNoStalenessPlanner(UWTIGPlanner):
    """Ablation: uncertainty + temporal + coverage, no persistent-monitoring
    staleness/latency term (Alamdari, Fata & Smith 2014)."""
    name = "uwtig_no_staleness"
    w_staleness = 0.0

class UWTIGNoLookaheadPlanner(UWTIGPlanner):
    """Ablation: pure 1-step greedy -- this project's original UW-TIG,
    before the Bircher-et-al./GATSBI-style receding-horizon lookahead."""
    name = "uwtig_no_lookahead"
    lookahead_discount = 0.0

class UWTIGNoCoverageFirstPlanner(UWTIGPlanner):
    """Ablation: no coverage-guarantee phase -- reproduces the
    recall/coverage-limited behavior UW-TIG had before this was added
    (coverage_frac stuck at 3/9 walls regardless of the other weights,
    since that ceiling comes from the cost-normalized-greedy formulation
    itself, not from the uncertainty/temporal/staleness terms -- see
    UWTIGPlanner's docstring)."""
    name = "uwtig_no_coverage_first"
    coverage_first = False
```

Appendix D - Localisation onto a wall (geometry.py)

```
def wall_plane(wall):
    """(point_on_plane, normal, u_axis, v_axis) for a WALL_SEGMENTS-style
    dict {x, y, yaw, width} (see sim_house.WALL_SEGMENTS), matching
    scene/wall_quad.obj's local-XZ-plane-facing-(-y) convention exactly:
    world = (x,y,Z_CENTER) + lx*(width/2)*u_axis + lz*(WALL_HEIGHT/2)*v_axis,
    normal = (sin(yaw), -cos(yaw), 0)."""
    yaw_rad = math.radians(wall["yaw"])
    normal = np.array([math.sin(yaw_rad), -math.cos(yaw_rad), 0.0])
    u_axis = np.array([math.cos(yaw_rad), math.sin(yaw_rad), 0.0])  # local +x -> world
    v_axis = np.array([0.0, 0.0, 1.0])  # local +z -> world, always vertical
    z_center = wall.get("z_center", 1.2)
    point = np.array([wall["x"], wall["y"], z_center])
    return point, normal, u_axis, v_axis

def ray_plane_intersect(origin, direction, plane_point, plane_normal):
    """Returns world xyz of intersection, or None if ray is parallel/behind."""
    denom = np.dot(direction, plane_normal)
    if abs(denom) < 1e-8:
        return None
    t = np.dot(plane_point - origin, plane_normal) / denom
    if t < 0:
        return None
    return origin + t * direction

def localize_on_wall(pixel_xy, drone_pos, drone_quat, wall, img_w, img_h, cam_offset_z=0.0,
                     within_extent=True):
    """Full pipeline: pixel -> ray -> intersection with the given wall's
    plane -> world xyz. Returns None if the ray misses the wall plane, or --
```

```
    when within_extent -- if the hit lies outside the wall's physical
    rectangle: that pixel shows some other surface (an adjacent wall, the
    ground, a prop) that merely shares the frame, not the wall being
    inspected. Without this check, 51-75% of all false positives on dev seeds
    were such misattributions, versus 1 of 1,369 true positives
    (experiments/diagnose_detection.py notes, RESULTS.md)."""
    view, proj = drone_camera_matrices(drone_pos, drone_quat, cam_offset_z=cam_offset_z)
    origin, direction = pixel_ray(pixel_xy[0], pixel_xy[1], view, proj, img_w, img_h)
    point, normal, u_axis, v_axis = wall_plane(wall)
    hit = ray_plane_intersect(origin, direction, point, normal)
    if hit is None or not within_extent:
        return hit
    d = hit - point
    if (abs(d @ u_axis) > wall["width"] / 2 + ON_WALL_MARGIN_M
            or abs(d @ v_axis) > WALL_HEIGHT / 2 + ON_WALL_MARGIN_M):
        return None
    return hit
```

Appendix E - Detector with TTA uncertainty (perception/ml_detector.py)

```
def _augment(bgr, rng):
    bright = cv2.convertScaleAbs(bgr, alpha=rng.uniform(0.8, 1.2), beta=rng.uniform(-20, 20))
    noisy = np.clip(bgr.astype(np.float32) + rng.normal(0, 10, bgr.shape), 0, 255).astype(np.uint8)
    blurred = cv2.GaussianBlur(bgr, (3, 3), 0)
    bright_blur = cv2.convertScaleAbs(cv2.GaussianBlur(bgr, (3, 3), 0),
                                      alpha=rng.uniform(0.85, 1.15), beta=rng.uniform(-15, 15))
    return [bright, noisy, blurred, bright_blur]

def square_pixels(bgr):
    """The drone camera (gym-pybullet-drones BaseAviary, mirrored in
    geometry.drone_camera_matrices) renders a 60 deg x 60 deg field of view
    into a non-square image (e.g. 320x240), so every frame is stretched
    horizontally by width/height (1.33x). The detector was trained on real,
    undistorted photos -- resample to square pixels (W x W) before inference.
    Returns (frame, y_scale) where y_scale maps detected y back to the
    original frame's coordinates, which geometry.py's localization expects."""
    h, w = bgr.shape[:2]
    if h == w:
        return bgr, 1.0
    return cv2.resize(bgr, (w, w), interpolation=cv2.INTER_LINEAR), h / w

def detect_with_uncertainty(bgr, weights=DEFAULT_WEIGHTS, conf_thresh=CONF_THRESH, seed=None, imgsz=320):
    model = _get_model(weights)
    rng = np.random.RandomState(seed) if seed is not None else np.random

    frame, y_scale = square_pixels(bgr)
    variants = [frame] + _augment(frame, rng)
    preds = [model.predict(v, conf=0.1, verbose=False, imgsz=imgsz)[0] for v in variants]

    canonical = preds[0]
    results = []
    for box in canonical.boxes:
        conf = float(box.conf.item())
        if conf < conf_thresh:
            continue
        cls_id = int(box.cls.item())
        x1, y1, x2, y2 = box.xyxy[0].tolist()
        bbox = (int(round(x1)), int(round(y1 * y_scale)), int(round(x2 - x1)), int(round((y2 - y1) * y_scale)))

        confs = [conf]
        for pred in preds[1:]:
            best_iou, best_conf = 0.0, 0.0
            for b2 in pred.boxes:
                if int(b2.cls.item()) != cls_id:
                    continue
                iou = _iou_xyxy((x1, y1, x2, y2), b2.xyxy[0].tolist())
                if iou > best_iou:
                    best_iou, best_conf = iou, float(b2.conf.item())
            confs.append(best_conf if best_iou > IOU_MATCH_THRESH else 0.0)

        results.append({
            "bbox": bbox, "label": CLASS_NAMES[cls_id], "confidence": conf,
            "uncertainty": float(np.std(confs)),
        })
    return results
```

Appendix F - One mission (experiments/mission.py)

```
def run_mission(planner_name, scenario, mission_index, detector, mem, rng,
```

```
            budget=10, gui=False, seed=0):
    """One mission: `scenario` must already have `assignment`/ground truth
    ready (see experiments/scenarios.py); ground truth for THIS mission
    index is scenario.ground_truth(mission_index). Returns a dict log.

    `seed` identifies this scenario+planner+seed run for memory isolation
    (see _seed_priors_from_memory) -- pass the actual experiment seed, not
    a fresh random draw, so all mission_index calls in the same run share
    one namespace and different (planner, seed) runs never collide."""
    wall_names = [w["name"] for w in WALL_SEGMENTS]
    run_key = f"{scenario.name}|{planner_name}|{seed}"
    viewpoints, dist = build_viewpoints(WALL_SEGMENTS)
    planner = PLANNER_REGISTRY[planner_name](viewpoints, dist, rng)
    belief = MissionBelief(WALL_SEGMENTS)

    mission_id = mem.create_mission(scenario.name, planner_name, seed, mission_index)
    if mission_index > 1:
        belief.seed_temporal_priors(_seed_priors_from_memory(mem, wall_names, run_key))

    house = KinematicHouse(gui=gui)
    with tempfile.TemporaryDirectory() as tmp_dir:
        gt = scenario.setup_mission(house, mission_index, tmp_dir)

        steps = []
        current_id = None
        total_dist = 0.0
        wall_detections = {w: [] for w in wall_names}

        for step_i in range(budget):
            vp = planner.select_next(belief, current_id)
            cost = dist[(current_id, vp.id)] if current_id else 0.0
            total_dist += cost

            bgr, pos, quat = house.capture(vp)
            wall_dict = house.wall_by_name[vp.wall]
            dets = detector(bgr)

            step_record = {"step": step_i, "viewpoint": vp.id, "wall": vp.wall,
                           "cost": cost, "cum_dist": total_dist, "detections": []}
            for det in dets:
                x, y, w, h = det["bbox"]
                center_px = (x + w / 2, y + h / 2)
                world_xyz = geo.localize_on_wall(center_px, pos, quat, wall_dict, IMG_W, IMG_H)
                if world_xyz is None:
                    continue
                corners = [geo.localize_on_wall((x, y), pos, quat, wall_dict, IMG_W, IMG_H, within_extent=False),
                           geo.localize_on_wall((x + w, y + h), pos, quat, wall_dict, IMG_W, IMG_H, within_extent=Fals
                corners = [c for c in corners if c is not None]
                size_m = float(np.linalg.norm(corners[0] - corners[1])) if len(corners) == 2 else 0.0

                crop = bgr[max(0, y):y + h, max(0, x):x + w]
                wall_key = f"{run_key}::{vp.wall}"
                det_id, defect_id, is_new, growth = mem.match_or_create_defect(
                    mission_id, wall_key, det["label"], det["confidence"], det["uncertainty"],
                    (x, y, w, h), tuple(world_xyz), size_m, crop_bgr=crop)

                belief.record_detection(vp.wall, world_xyz, det["uncertainty"], growth, wall_dict)
                wall_detections[vp.wall].append(det["label"])
                step_record["detections"].append({
                    "label": det["label"], "confidence": det["confidence"],
                    "uncertainty": det["uncertainty"], "world_xyz": list(world_xyz),
                    "size_m": size_m, "defect_id": defect_id, "is_new": is_new, "growth": growth,
                })

            belief.record_visit(vp)
            steps.append(step_record)
            current_id = vp.id

        mean_entropy = float(np.mean([belief.coverage_entropy(w["name"], c)
                                      for w in WALL_SEGMENTS for c in range(n_cells_for_wall(w))]))
    house.close()

    return {
        "planner": planner_name, "scenario": scenario.name, "mission_index": mission_index,
        "steps": steps, "total_dist": total_dist, "n_viewpoints": budget,
        "wall_detections": wall_detections, "ground_truth": gt,
        "final_mean_entropy": mean_entropy,
    }
```

Appendix G - Kinematic capture (experiments/kinematic_capture.py)

```
class KinematicHouse:
    def __init__(self, gui=False):
        self.client = p.connect(p.GUI if gui else p.DIRECT)
        p.setAdditionalSearchPath(pybullet_data.getDataPath(), physicsClientId=self.client)
        plane_id = p.loadURDF("plane.urdf", physicsClientId=self.client)
        retexture_ground(self.client, plane_id)
        self.wall_bodies = build_house(self.client)  # {wall_name: body_id}
        self.wall_by_name = {w["name"]: w for w in WALL_SEGMENTS}

    def set_wall_texture(self, wall_name, texture_path):
        tex_id = p.loadTexture(texture_path, physicsClientId=self.client)
        p.changeVisualShape(self.wall_bodies[wall_name], -1, textureUniqueId=tex_id,
                            physicsClientId=self.client)

    def capture(self, viewpoint):
        """Returns (bgr_uint8, drone_pos, drone_quat) at the given Viewpoint."""
        quat = p.getQuaternionFromEuler([0, 0, math.radians(viewpoint.yaw_deg)])
        view, proj = geo.drone_camera_matrices(viewpoint.pos, quat)
        view_flat = view.flatten(order="F").tolist()
        proj_flat = proj.flatten(order="F").tolist()
        _, _, rgba, _, _ = p.getCameraImage(
            width=int(IMG_RES[0]), height=int(IMG_RES[1]),
            viewMatrix=view_flat, projectionMatrix=proj_flat,
            shadow=1, lightDirection=[0.6, -0.4, 1.0],
            renderer=p.ER_TINY_RENDERER, physicsClientId=self.client,
        )
        rgb = np.reshape(rgba, (int(IMG_RES[1]), int(IMG_RES[0]), 4))[:, :, :3].astype(np.uint8)
        bgr = rgb[:, :, ::-1].copy()
        return bgr, viewpoint.pos, quat

    def close(self):
        p.disconnect(physicsClientId=self.client)
```

Appendix H - Viewpoint graph (planning/viewpoints.py)

```
def build_viewpoints(wall_segments):
    """Returns (viewpoints: list[Viewpoint], dist: dict[(id,id)] -> meters)."""
    viewpoints = []
    for wall in wall_segments:
        point, normal, u_axis, v_axis = geo.wall_plane(wall)
        half_width = wall["width"] / 2
        for standoff in STANDOFFS:
            for frac in LATERAL_FRACS:
                lateral = frac * half_width
                pos = point + standoff * normal + lateral * u_axis
                pos[2] = FLIGHT_Z
                approach_yaw = wall["yaw"] + 90
                vid = f"{wall['name']}|s{standoff:.1f}|l{frac:+.2f}"

                visible_half_span = standoff * math.tan(HALF_FOV_RAD)
                lo_u, hi_u = lateral - visible_half_span, lateral + visible_half_span
                n_cells = n_cells_for_wall(wall)
                cell_w = wall["width"] / n_cells
                visible_cells = []
                for c in range(n_cells):
                    cell_lo = -half_width + c * cell_w
                    cell_hi = cell_lo + cell_w
                    if cell_hi > lo_u and cell_lo < hi_u:
                        visible_cells.append(c)

                viewpoints.append(Viewpoint(
                    id=vid, wall=wall["name"], pos=pos, yaw_deg=approach_yaw,
                    standoff=standoff, lateral=lateral, visible_cells=visible_cells,
                ))

    dist = {}
    for a in viewpoints:
        for b in viewpoints:
            dist[(a.id, b.id)] = float(np.linalg.norm(a.pos - b.pos))
    return viewpoints, dist
```